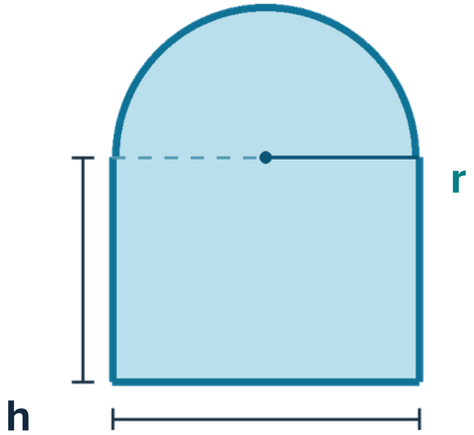
PROBLEM 01 · MAXIMIZATION

The Norman Window: Define the Geometry



2r

Rectangle + semicircle

GOAL: MAXIMUM AREA

For a fixed perimeter P, what window shape allows the most light?

The outside boundary includes:

2 vertical sides (2h) + bottom (2r)
+ semicircular arc (πr)

$$P = 2h + (2 + \pi)r$$

From the Area Formula to the Critical Radius

1 Solve for the rectangular height

$$h = [P - (2 + \pi)r] / 2$$

2 Area of the rectangle + semicircle

$$A = 2rh + \pi r^2/2$$

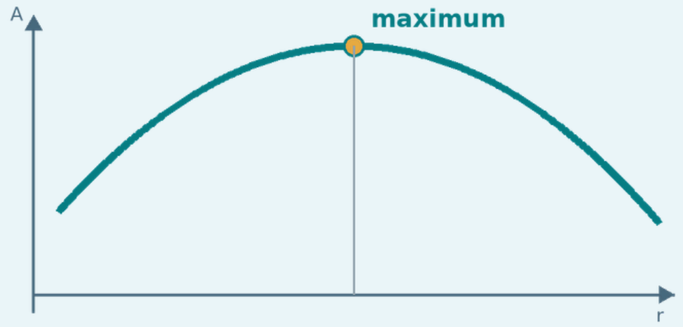
$$A(r) = Pr - (4 + \pi)r^2/2$$

3 Differentiate, then set the result to zero

$$A'(r) = P - (4 + \pi)r = 0$$

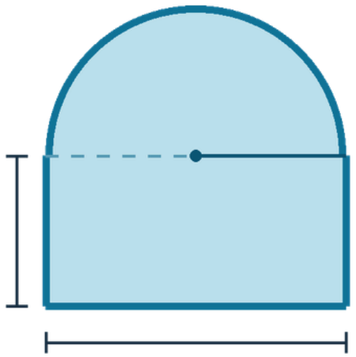
CRITICAL RADIUS

$$r = P / (4 + \pi)$$



Concave-down area function

The Optimal Norman Window



$$h = r$$

RECTANGLE HEIGHT = SEMICIRCLE RADIUS

$$h = r = P / (4 + \pi)$$

Second-derivative check

$$A''(r) = -(4 + \pi) < 0 \rightarrow \text{MAXIMUM}$$

MAXIMUM WINDOW AREA

$$A_{max} = P^2 / [2(4 + \pi)]$$

Width = total height = 2r

PROBLEM 02 · MINIMIZATION

Two Numbers With a Fixed Sum

Given $x + y = K$, find the smallest possible value of $S = x^2 + y^2$.

THE CONSTRAINT

$$x + y = K$$



Illustration for $K > 0$

1 Express one number in terms of the other

$$y = K - x$$

2 Substitute into the objective function

$$S(x) = x^2 + (K - x)^2$$

$$S(x) = 2x^2 - 2Kx + K^2$$

Differentiate and Find the Minimum

1 Differentiate S(x)

$$S'(x) = 4x - 2K$$

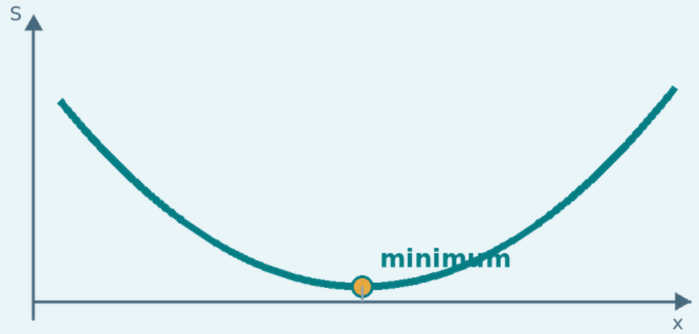
2 Set the derivative to zero

$$4x - 2K = 0 \rightarrow x = K/2$$

$$y = K - x = K/2$$

3 Check the second derivative

$$S''(x) = 4 > 0 \rightarrow \text{MINIMUM}$$



Concave upward → minimum

MINIMUM SUM OF SQUARES

$$S_{min} = K^2 / 2$$

Worked Example and Key Takeaways

EXAMPLE: K = 10

Compare possible pairs

Numbers	$x^2 + y^2$
1 and 9	82
2 and 8	68
3 and 7	58
4 and 6	52
5 and 5	**50 ✓**

PROBLEM 1 — MAXIMUM

Norman window: $h = r$

$$A_{max} = P^2 / [2(4 + \pi)]$$

PROBLEM 2 — MINIMUM

$$x = y = K / 2$$

$$S_{min} = K^2 / 2$$

$f''(x) < 0$: maximum | $f''(x) > 0$: minimum